

WASTEWISE SCAN · REQUIREMENTS

WasteWise Scan requirements

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DOCUMENT CONTROL

Document ID	WSC-REQ-001	Version	0.3
Status	Draft	Date	2026-09-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wastewise-scan		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 status from WSC-CAL-001; R12 redefined per WSC-DDR-001 D2; R13 points to the draft record in WSC-DDR-002
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

WasteWise Scan requirements

These are the requirements for the TRL 3 design. Status is from the calculation note WSC-CAL-001 (tags in brackets). **R4 and R7 are not met** on paper, R1 and R13 are at risk, and R3 and R10 cannot be verified at TRL 3. Targets stay proposals for co-design with users; the only target changed at this revision is R12, redefined by Amish's decision of 2026-09-25 (WSC-DDR-001, D2).

Table 1. Requirements.

ID	Requirement	Target	Status at TRL 3 (WSC-CAL-001)	Verification (TRL 3 or later)
R1	Identify common resins	PET, HDPE, PP, PS and PVC, plus "other" and "unknown"; 95 % or more correct on clean, dry, non-black items of 1 mm wall or more	At risk. The 1,650 nm band reaches only the short edge of the 1,650 to 1,750 nm region on standard InGaAs (centroid about 1,628 nm) [B2]; signal-to-noise ratio of 450 or more in shade [C4]	Band study on published reference spectra; later a labeled item set of 200 or more
R2	Fast result	1.5 s or less from button press to result	Met: about 0.33 s [E2]	Timing budget
R3	Fail safe on unreadable items	Black, dark, wet or very thin items read "unknown"; wrong resin shown for 2 % or fewer of all scans	Not verifiable at TRL 3. Black items fall well below the 0.08 reflectance threshold [C6]; the wrong-result rate needs item data	Confidence threshold study; later item test
R4	Work in sunlight	Same result indoors and in direct sun up to 100 klx when the shroud is pressed to the item	Not met for clear and translucent items: light through the item is about 900 times the weakest signal and hand movement leaves an error several times the signal [D3]. Met by estimate for opaque items [D5]	Stray light estimate
R5	Last a shift	8 h or more at 4 scans per minute on one charge; USB-C charging	Met: about 22 h [F2]	Power budget
R6	One-handed and light	250 g or less; body no larger than 170 x 65 x 40 mm (6.7 x 2.6 x 1.6 in) plus shroud	Met: about 202 g, 222 g with cap; 160 x 62 x 34 mm [A1, A3]	Parametric model, then weighing
R7	Readable by anyone	Result as resin code, name, color and icon, legible at arm's length in direct sun; local language; no reading needed to use	Not met in direct sun: about 1.6:1 contrast against about 3:1 needed; about 5.2:1 when shaded [I1, I2]	Contrast estimate; design review with users
R8	Local price grade	Shows a price grade from a table the user or cooperative edits on the device; works offline	Met by design	Design review
R9	Log scans	10,000 or more scans stored on the device with time, result, confidence and spectrum; export as CSV over USB; no cloud needed	Met: about 131,000 scans in an 8 MB partition [G1]	Storage calculation
R10	Rugged	Survives 1.2 m (4 ft) drop onto concrete; IP54 target; 0 to 45 °C	Not verifiable at TRL 3. Corner drop about 1,200 g; the cradle must hold the cell against about 550 N [J1, J2]	Design review, later drop and spray tests
R11	Stay calibrated	White reference check in 10 s or less using the calibration	Met by design; 5 % drift after about 12.5 °C of temperature change [J3]	Design review

ID	Requirement	Target	Status at TRL 3 (WSC-CAL-001)	Verification (TRL 3 or later)
		cap; warns when a check is overdue or drift exceeds 5 %		
R12	Low cost	\$150 or less per unit is the volume target; the first prototype is accepted at about \$163 in parts; no custom PCB for the first build	Met for the prototype: \$163.00 [K1]. The volume target cannot be shown without volume quotes	Priced BOM
R13	Work with WasteWise-ml	Low-confidence scans hand off to WasteWise-ml; scan records use a shared, documented format	At risk. Hand-off designed; record format drafted in WSC-DDR-002, not yet agreed with WasteWise-ml	Interface note with WasteWise-ml
R14	Eye-safe illumination	LED emission in the IEC 62471 exempt group; LEDs pulse only during a scan	Met by estimate: 3.9 W/m ² at 200 mm against 100 W/m ² ; retinal 3.8 % of the limit [H2, H3]	Emission estimate, then datasheet values

■ Assumptions

- "Clean and dry" means rinsed or wiped items without labels over the scan spot. Real waste is dirtier; the classifier must reject rather than guess.
- About 10 to 15 % of items in a mixed post-consumer stream are black or very dark (estimate; varies strongly by region and should be measured with users).
- The first prototype's accepted cost of about \$163 covers one scanner with its cap; the \$150 volume target assumes quantity prices for LEDs and the photodiode (WSC-DDR-001, D2).
- Four scans per minute is a busy spot-check pace, not every item: most items are sorted by eye and the scanner settles doubtful ones.
- Resin prices vary by city and season; the device stores grades (for example A, B, C, reject) and a local price list, not global prices.